Guía3 - Ej8

Ecuaciones de equilibrio diferenciales

$$1)\ \frac{\partial \sigma_x}{\partial x}+\frac{\partial \tau_{xy}}{\partial y}+X=0,\ 2)\ \frac{\partial \tau_{xy}}{\partial x}+\frac{\partial \sigma_y}{\partial y}+Y=0$$

$\Phi(x,y)$ → Función de tensión de Airy

→ Fuerzas de cuerpo / Fuerza por unidad de volumen

$$F=\begin{bmatrix} X \\ Y \end{bmatrix}$$ → son constantes en (x,y)

$$\boxed{\sigma_x=\frac{\partial^2\Phi}{\partial y^2},\quad \sigma_y=\frac{\partial^2\Phi}{\partial x^2},\quad \tau_{xy}=-\frac{\partial^2\Phi}{\partial x\,\partial y}-Xy-Yx}$$

→ coordenadas

$$\begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{xy} & \sigma_y \end{bmatrix}$$

1)
$$\frac{\partial^3\Phi}{\partial x\partial y^2}+\frac{\partial}{\partial y}\left(-\frac{\partial^2\Phi}{\partial x\partial y}-Xy-Yx\right)+X=0$$

$$\cancel{\frac{\partial^3\Phi}{\partial x\partial y^2}}-\cancel{\frac{\partial^3\Phi}{\partial x\partial y^2}}-\cancel{X}+\cancel{X}=0$$

2)
$$-\cancel{\frac{\partial^3\Phi}{\partial x^2\partial y}}-\cancel{Y}+\cancel{\frac{\partial^3\Phi}{\partial y\partial x^2}}+\cancel{Y}=0$$

→ En equilibrio

a. $\Phi(x,y)=ax^2+bxy+cy^2$

$$\sigma_x=\frac{\partial^2\Phi}{\partial y^2},\quad \sigma_y=\frac{\partial^2\Phi}{\partial x^2},\quad \tau_{xy}=-\frac{\partial^2\Phi}{\partial x\,\partial y}-Xy-Yx$$

i. $a\neq 0,\quad b=c=0$ → Estados tensionales distintos

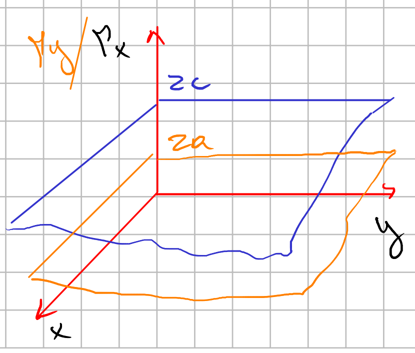
ii. $b\neq 0,\quad a=c=0$ →

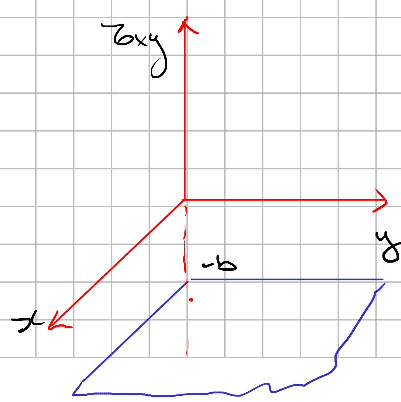
iii. $c\neq 0,\quad a=b=0$ →

$$\frac{\partial\Phi}{\partial x}=2ax+by \qquad \frac{\partial\Phi}{\partial y}=bx+2cy$$

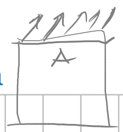
$$\begin{cases} \sigma_x=2c=\frac{\partial}{\partial y}(bx+2cy)=2c \\ \sigma_y=2a \\ \tau_{xy}=-b-Xy-Yx \end{cases}$$

$X=0$
$Y=0$ $\Rightarrow$ $\underline{\underline{\sigma}}(x,y)=\begin{bmatrix} 2c & -b \\ -b & 2a \end{bmatrix}$

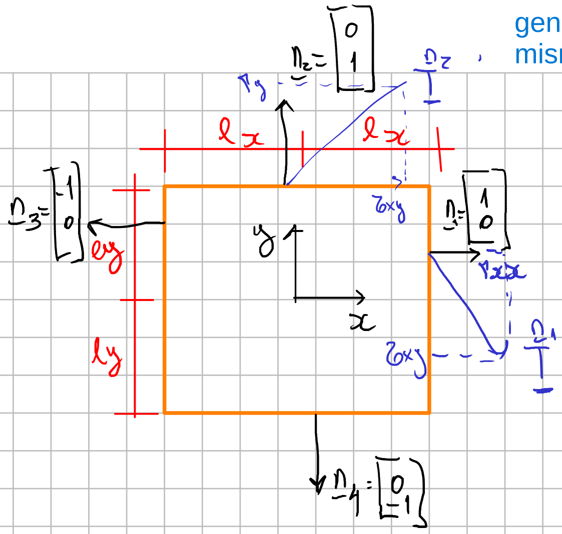




general para la misma geometria



$F = \int_A T \, dA$



$$\underline{T}(x,y) = \underline{\underline{\sigma}}(x,y)\cdot\underline{n} = \begin{bmatrix}\sigma_{xx} & \tau_{xy}\\ \tau_{xy} & \sigma_{yy}\end{bmatrix}\cdot\begin{bmatrix}n_x\\ n_y\end{bmatrix}$$

$$\underline{T}^{n_1} = \begin{bmatrix}\sigma_{xx}(x,y) & \tau_{xy}(x,y)\\ \tau_{xy}(x,y) & \sigma_{yy}(x,y)\end{bmatrix}\cdot\begin{bmatrix}1\\ 0\end{bmatrix} = \begin{bmatrix}\sigma_{xx}(\underline{l_x},y)\\ \tau_{xy}(\underline{l_x},y)\end{bmatrix}$$

$$\underline{T}^{n_2} = \begin{bmatrix}\tau_{xy}(x,\underline{l_y})\\ \sigma_{yy}(x,\underline{l_y})\end{bmatrix}$$

$$\underline{T}^{n_3} = -\begin{bmatrix}\sigma_{xx}(-\underline{l_x},y)\\ \tau_{xy}(-\underline{l_x},y)\end{bmatrix} \qquad \underline{T}^{n_4} = -\begin{bmatrix}\tau_{xy}(x,-\underline{l_y})\\ \sigma_{yy}(x,-\underline{l_y})\end{bmatrix}$$

a) Caso i $\quad a \neq 0,\ b = 0,\ c = 0$

$$\underline{F} = \begin{bmatrix}X\\ Y\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$ fuerzas masicas

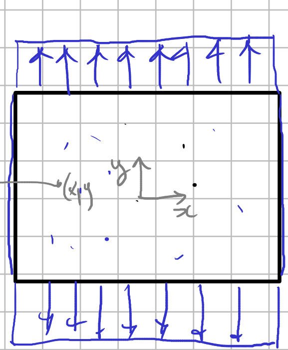
$$\left.\begin{aligned}\sigma_x &= 2c = 0\\ \sigma_y &= 2a = 2a\\ \tau_{xy} &= -b = 0\end{aligned}\right\} \Rightarrow \underline{\underline{\sigma}}(x,y) = \begin{bmatrix}0 & 0\\ 0 & 2a\end{bmatrix}$$

$$\underline{T}^{n_1} = \begin{bmatrix}0\\ 0\end{bmatrix} \qquad \underline{T}^{n_2} = \begin{bmatrix}0\\ \underline{2a}\end{bmatrix} \qquad \underline{T}^{n_3} = \begin{bmatrix}0\\ 0\end{bmatrix} \qquad \underline{T}^{n_4} = \begin{bmatrix}0\\ -2a\end{bmatrix}$$

$a > 0 \Rightarrow$ traccionado

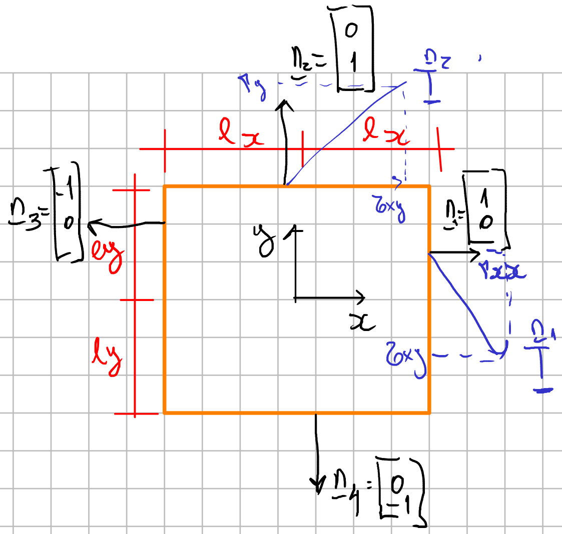
$\underline{T}^{n_2} = 2a$

$\begin{bmatrix}0 & 0\\ 0 & 2a\end{bmatrix} = \underline{\underline{\sigma}}$



$\underline{T}^{n_4} = -\underline{T}^{n_2} = -2a$

Esfuerzos axiales en dir. y
$\downarrow$
$\rightarrow$ Traccionado



$$\underline{T}(x,y) = \underline{\underline{\sigma}}(x,y)\cdot \underline{n} = \begin{bmatrix}\sigma_{xx} & \tau_{xy}\\ \tau_{xy} & \sigma_{yy}\end{bmatrix}\cdot\begin{bmatrix}n_x\\ n_y\end{bmatrix}$$

$$\overset{n_1}{\underline{T}} = \begin{bmatrix}\sigma_{xx}(x,y) & \tau_{xy}(x,y)\\ \tau_{xy}(x,y) & \sigma_{yy}(x,y)\end{bmatrix}\cdot\begin{bmatrix}1\\ 0\end{bmatrix} = \begin{bmatrix}\sigma_{xx}(\underline{l_x},y)\\ \tau_{xy}(\underline{l_x},y)\end{bmatrix}$$

$$\overset{n_2}{\underline{T}} = \begin{bmatrix}\tau_{xy}(x,\underline{l_y})\\ \sigma_{yy}(x,\underline{l_y})\end{bmatrix}$$

$$\overset{n_3}{\underline{T}} = -\begin{bmatrix}\sigma_{xx}(\underline{-l_x},y)\\ \tau_{xy}(\underline{-l_x},y)\end{bmatrix} \qquad \overset{n_4}{\underline{T}} = -\begin{bmatrix}\tau_{xy}(x,\underline{-l_y})\\ \sigma_{yy}(x,\underline{-l_y})\end{bmatrix}$$

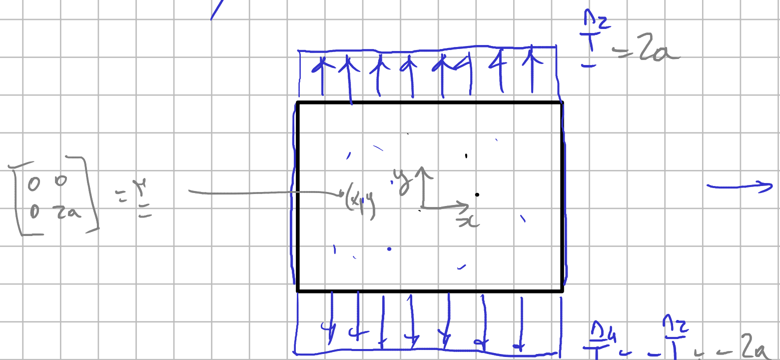
a) Caso i $\quad a \neq 0,\ b=0,\ c=0 \qquad \underline{F} = \begin{bmatrix}X\\ Y\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$

campo de tensiones

$$\left.\begin{aligned}\sigma_x &= 2c = 0\\ \sigma_y &= 2a = 2a\\ \tau_{xy} &= -b = 0\end{aligned}\right\} \Rightarrow \underline{\underline{\sigma}}(x,y) = \begin{bmatrix}0 & 0\\ 0 & 2a\end{bmatrix}$$

$$\overset{n_1}{\underline{T}} = \begin{bmatrix}0\\ 0\end{bmatrix} \qquad \overset{n_2}{\underline{T}} = \begin{bmatrix}0\\ \underline{2a}\end{bmatrix} \qquad \overset{n_3}{\underline{T}} = \begin{bmatrix}0\\ 0\end{bmatrix} \qquad \overset{n_4}{\underline{T}} = \begin{bmatrix}0\\ -2a\end{bmatrix}$$

$a > 0 \Rightarrow$ traccionado



Esfuerzos axiales en dir. y

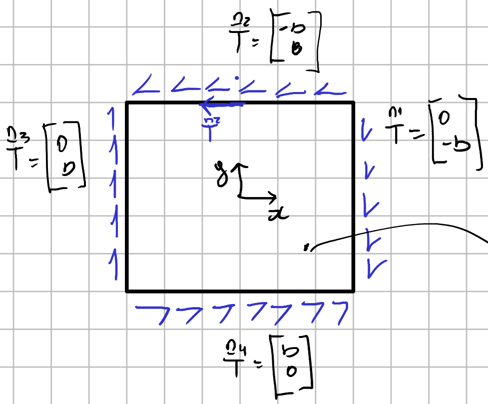
↓

→ Traccionada

caso ii) $b \neq 0$, $a = c = 0$ $\Rightarrow \underline{\underline{\sigma}} = \begin{bmatrix} 0 & -b \\ -b & 0 \end{bmatrix}$

$\overset{n_1}{T} = \begin{bmatrix} 0 \\ -b \end{bmatrix}$, $\overset{n_3}{T} = \begin{bmatrix} 0 \\ b \end{bmatrix}$, $\overset{n_2}{T} = \begin{bmatrix} -b \\ 0 \end{bmatrix}$, $\overset{n_4}{T} = \begin{bmatrix} b \\ 0 \end{bmatrix}$

$b > 0$

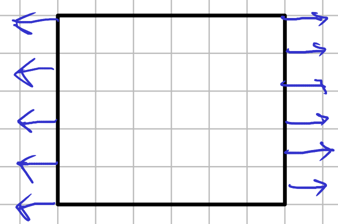


Esfuerzo de corte puro

$\underline{\underline{\sigma}} = \begin{bmatrix} 0 & -b \\ -b & 0 \end{bmatrix}$

caso iii) $c \neq 0$, $a = 0$, $b = 0$ $\Rightarrow$ $\sigma = \begin{bmatrix} 2c & 0 \\ 0 & 0 \end{bmatrix}$

$c > 0$



Esf. axial en direc. x

b. $\Phi(x,y) = ax^3 + bx^2y + cxy^2 + dy^3$

i. $d \neq 0, \quad a = b = c = 0$

ii. $a \neq 0, \quad b = c = d = 0$

iii. $b \neq 0, \quad a = c = d = 0$

$$\sigma_x = \frac{\partial^2 \Phi}{\partial y^2}, \quad \sigma_y = \frac{\partial^2 \Phi}{\partial x^2}, \quad \tau_{xy} = -\frac{\partial^2 \Phi}{\partial x \partial y} - \underset{0}{Xy} - \underset{0}{Yx}$$

$$\frac{\partial \Phi}{\partial x} = 3ax^2 + 2bxy + cy^2$$

$$\frac{\partial \Phi}{\partial y} = bx^2 + 2cxy + 3dy^2$$

$$\underline{\underline{\sigma}} = \begin{cases} \sigma_x = 2cx + 6dy \\ \sigma_y = 6ax + 2by \end{cases} \qquad \tau_{xy} = -2bx - 2cy$$

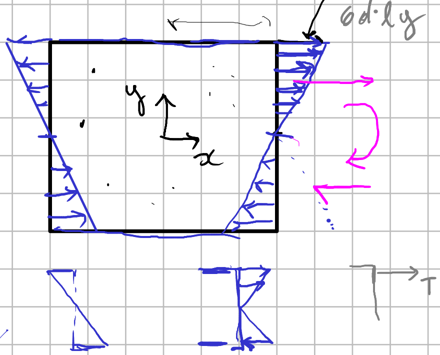
i. $d \neq 0, \quad a = b = c = 0$

$\sigma_{xx}(x,y) = 6dy$

$$\underline{\sigma}(x,y) = \begin{bmatrix} 6dy & 0 \\ 0 & 0 \end{bmatrix}$$

$$\underline{T}^{n_1} = \begin{bmatrix} 6dy \\ 0 \end{bmatrix} \qquad d > 0$$

$6d \cdot ly$

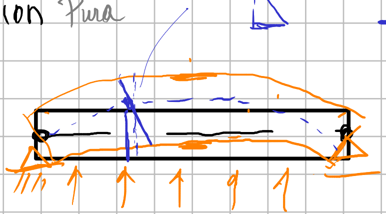


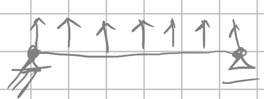
$$\underline{T}^{n_2} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\underline{T}^{n_3} = \begin{bmatrix} -6dy \\ 0 \end{bmatrix}$$

$$\underline{T}^{n_4} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Flexión Pura





caso iii) $\quad b \neq 0, \quad a = c = d = 0$

$$\underline{\sigma} \begin{cases} \sigma_x = 2cx + 6dy \\ \sigma_y = 6ax + 2by \end{cases} \qquad \tau_{xy} = -2bx - 2cy$$

$$\sigma_x = 0$$
$$\sigma_y = 2by$$
$$\tau_{xy} = -2bx$$

$$\sigma = \begin{bmatrix} 0 & -2bx \\ -2bx & 2by \end{bmatrix}$$

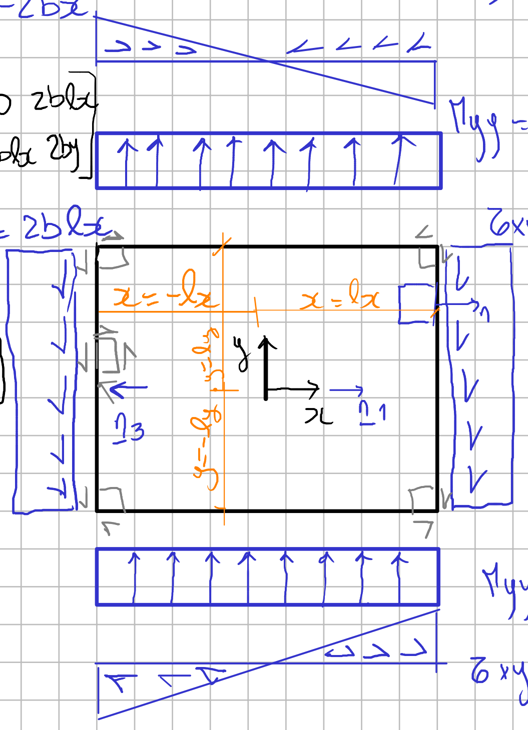
vector tensor de corte $\quad T_x$

$b > 0 \qquad \frac{n_2}{T} = \sigma \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2bx \\ 2by \end{bmatrix} = \begin{bmatrix} -2bx \\ 2b\,l_y \end{bmatrix}$

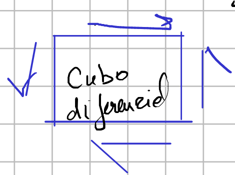
$T_x = \tau_{xy} = -2bx$

$$\underline{\sigma}(-l_x, y) = \begin{bmatrix} 0 & 2bl_x \\ 2bl_x & 2by \end{bmatrix}$$

$\tau_{xy} = -2b(-l_x) = 2bl_x$



$$\frac{n_1}{T} = \sigma \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -2bx \end{bmatrix} = \begin{bmatrix} 0 \\ -2bl_x \end{bmatrix}$$



$$\frac{n_3}{T} = \underline{\sigma} \cdot \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2bx \end{bmatrix} = \begin{bmatrix} 0 \\ -2bl_x \end{bmatrix}$$

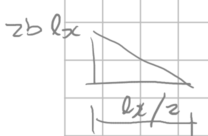
$$\frac{n_4}{T} = \sigma \cdot \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2bx \\ -2by \end{bmatrix} = \begin{bmatrix} 2bx \\ -2b\cdot(-l_y) \end{bmatrix} = \begin{bmatrix} 2bx \\ 2bl_y \end{bmatrix}$$

La estructura tiene que verificar el equilibrio global:

$$T = \begin{bmatrix} T_x \\ T_y \end{bmatrix}$$

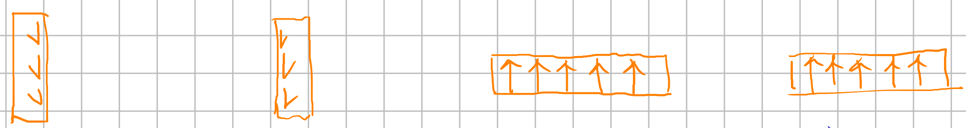
$$\Sigma F_x = 0 \Rightarrow \int_\Gamma T_x \, ds = 0 \Rightarrow$$

$\Gamma \to$ frontera



$$2b \cdot \cancel{l_x} \cdot \frac{l_x}{2} - 2b\cancel{l_x} \cdot \frac{l_x}{2} - \cancel{2bl_x} \cdot \frac{l_x}{2} + 2b\cancel{l_x} \cdot \frac{l_x}{2} = 0 \Rightarrow \text{B.C.}$$

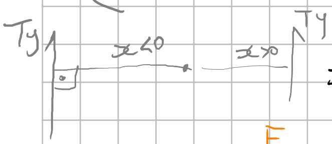
$$\Sigma F_y = 0 \Rightarrow \int_\Gamma T_y\, ds = 0$$



$$-2b\cancel{l_x}\,2l_y - 2b\cancel{l_x}.2l_y + 2b\cancel{l_y}.2l_x + 2b\cancel{l_y}.2l_x = 0 \Rightarrow \text{B.C.}$$

$$\Sigma M_z = 0 \Rightarrow \int_\Gamma (\underline{r} \times \underline{T}) \cdot \underline{k}\; ds$$

$$\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ x & y & 0 \\ T_x & T_y & 0 \end{vmatrix} = (xT_y - yT_x)\,\underline{k}$$

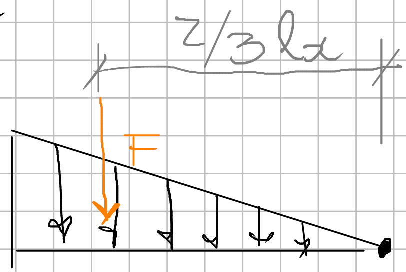


$$= \int_\Gamma \left(x.\overbrace{T_y\, ds}^{dF_y} - y.\overbrace{T_x\, ds}^{dF_x}\right) =$$

$$= \overbrace{2bl_y 2l_x}^{F}.\overbrace{0}^{d} + \overbrace{2bl_y.2l_x}^{F}.\overbrace{0}^{d} + \overbrace{2b\cancel{l_x.2l_y}}^{F}.\overbrace{l_x}^{d} - \overbrace{2b\cancel{l_x.2l_y}}^{F}.\overbrace{l_x}^{d} -$$

$$-\underbrace{2b\cancel{l_x.\tfrac{l_x}{2}}}_{F}.\underbrace{l_y}_{d} + \underbrace{2b\cancel{l_x.\tfrac{l_x}{2}}}_{F}.\underbrace{l_y}_{d} - \underbrace{2bl_x.\cancel{\tfrac{l_x}{2}}}_{F}.\underbrace{l_y}_{d} + \underbrace{2bl_x\cancel{\tfrac{l_x}{2}}}_{F}.\underbrace{l_y}_{d} =$$

$$= 0 \Rightarrow \text{B.C}$$



$$M_z = F.d = F.\,{}^{2}/_{3}\, l_x$$